

Bang and the Artefacts are Gone! Rapid Artefact Removal and Tissue Segmentation in Haematoxylin and Eosin Stained Biopsies

Benjamin Schreiber (✉ bas43@cam.ac.uk)

University of Cambridge

James Denholm

University of Cambridge

Florian Jaeckle

Lyzeum Ltd.

Mark Arends

University of Edinburgh

Kim Branson

GlaxoSmithKline (United Kingdom)

Carola-Bibiane Schönlieb

University of Cambridge

Elizabeth Soilleux

Lyzeum Ltd.

Article

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Posted Date: November 7th, 2023

DOI: <https://doi.org/10.21203/rs.3.rs-3503968/v1>

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Additional Declarations: Competing interest reported. The following authors are shareholders in Lyzeum Ltd.: Elizabeth Jane Soilleux and Carola-Bibiane Schönlieb

Bang and the Artefacts are Gone! Rapid Artefact Removal and Tissue Segmentation in Haematoxylin and Eosin Stained Biopsies

B. A. Schreiber^{1,2*}, J. Denholm^{1,2,3}, F. Jaeckle^{1,3}, M. J. Arends⁴, K. M. Branson⁵, C.-B. Schönlieb^{2,3}, E. J. Soilleux^{1,3*}

^{1*}Department of Pathology, University of Cambridge, Tennis Court Road, Cambridge, CB2 1QP, Cambridgeshire, United Kingdom.

²Department of Applied Mathematics and Theoretical Physics, University of Cambridge, Wilberforce Road, Cambridge, CB3 0WA, Cambridgeshire, United Kingdom.

³Lyzeum ltd, Cambridge, CB1 2LA, Cambridgeshire, United Kingdom.

⁴Edinburgh Pathology, Institute of Genetics & Cancer, University of Edinburgh, Crewe Road, Edinburgh, EH4 2XR, UK.

⁵Artificial Intelligence and Machine Learning, GlaxoSmithKline plc, Great West Road, Brentford, TW8 9GS, Middlesex, UK.

*Corresponding author(s). E-mail(s): bas43@cam.ac.uk; ejs17@cam.ac.uk;

Abstract

We present H&E Otsu thresholding, a scheme for rapidly detecting tissue in whole-slide images (WSIs) that eliminates a wide range of undesirable artefacts such as pen marks and scanning artefacts. Our method involves obtaining a bimodal representation of a low-magnification RGB overview image which enables simple Otsu thresholding to separate tissue from background and artefacts. We demonstrate our method on WSIs prepared from a wide range of institutions and WSI digital scanners, each containing substantial artefacts that cause other methods to fail. The beauty of our approach lies in its simplicity: manipulating RGB colour space and using Otsu thresholding allows for the rapid removal of artefacts and segmentation of tissue.

Keywords: Artefact Removal, Tissue Segmentation, Whole-Slide Imaging, Haematoxylin and Eosin, Machine Learning

1 Introduction

Otsu thresholding [1] is often used to identify tissue in a whole-slide image (WSI) of a haematoxylin and eosin (H&E)-stained biopsy (see Figure 1), including in popular histopathological analysis tools Histolab [2] and PyHist [3].

However, Otsu thresholding only successfully segments the tissue from the background when the tissue and background pixels are well-separated in a greyscale representation of the WSI. While this is often the case in artefact-free WSIs, WSIs often contain artefacts such as pen marks and dark scanning artefacts, which cause this assumption to fail,

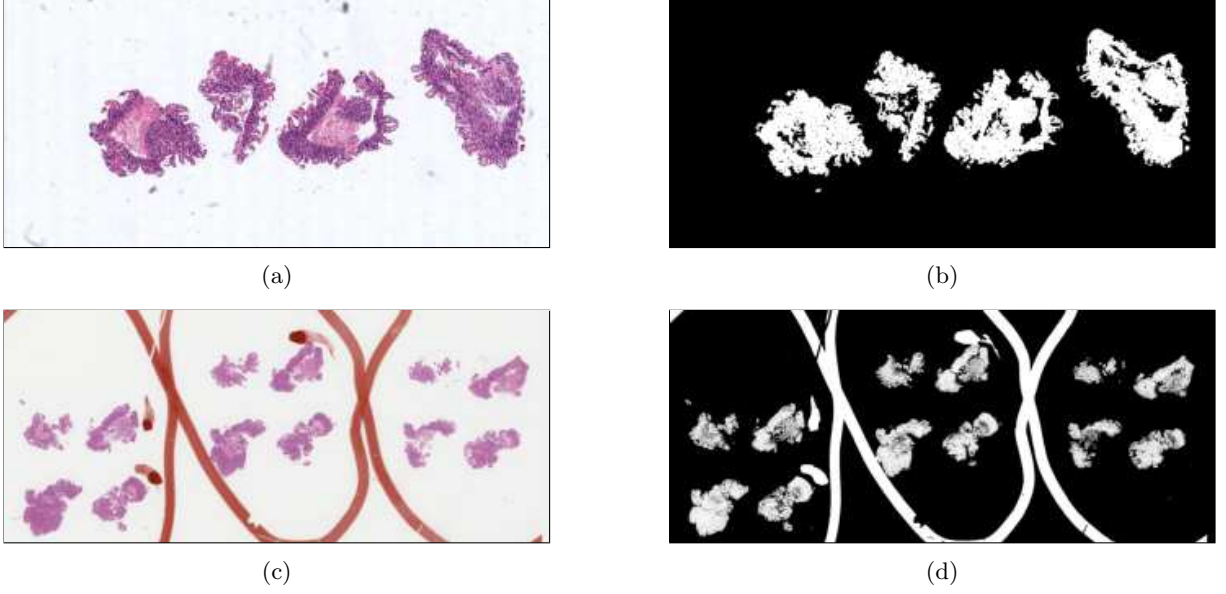


Fig. 1: (a) A WSI of a H&E-stained biopsy containing only minor, non-interfering artefacts. (b) The tissue segmentation provided by Otsu thresholding. The clear distinction between the intensities of the tissue and the rest of the WSI caused the Otsu threshold to lie between the maximum intensity of the tissue and the minimum intensity of background, allowing for a successful tissue segmentation. (c) A WSI of an H&E-stained biopsy containing orange pen marks splitting the tissue of different levels and used to identify three features of interest. (d) The result of segmenting the tissue using standard Otsu thresholding. The pen marks interfered with the Otsu threshold calculation, resulting in a tissue segmentation that contains tissue and pen marks.

thus resulting in artefacts wrongly identified as tissue, tissue rejected as background, or both (see Figure 1).

The exclusion of pen marks in particular is a crucial first step for any machine learning-based automated WSI analysis pipeline; pathologists often use pen marks to highlight areas of interest which, if observed by a machine learning algorithm, could result in deleterious bias, spurious classifications or even data leakage, thus reduce the confidence in the performance metrics and the generalizability of the algorithm [4].

In this paper, we propose a method that improves on Otsu thresholding as a tissue segmenting algorithm for H&E-stained biopsies by successfully segmenting tissue in the presence of artefacts and rejecting artefacts as background. We tested our method on WSIs of H&E-stained duodenal biopsies prepared at multiple different institutions, scanned using multiple different scanners, and containing a large range of artefacts of different types, shapes and colours.

2 Method

Our method improves on Otsu thresholding by selecting a representation of the WSI data that better separates H&E-stained tissue from background and artefacts. Given a three channel image $I = [I_R, I_B, I_G]$, the following representation of the data is calculated:

$$T = \text{ReLU}(I_R - I_G) \odot \text{ReLU}(I_B - I_G) \quad (1)$$

where $\text{ReLU}(x) = \max(x, 0)$ is the rectifier linear unit and $\odot$ is the Hadamard product, both of which act element-wise. Otsu thresholding is then used to separate tissue and non-tissue pixels [1]. Note that this calculation requires no parameter training or tuning. A Python implementation of this previously unreported algorithm can be found here https://gitlab.developers.cam.ac.uk/bas43/h_and_e_otsu_thresholding in accordance with the Guidelines for Authors

Submitting Code and Software presented in Nature Research <https://www.nature.com/nature-portfolio/editorial-policies/reporting-standard#reporting-requirements>.

All relevant guidelines were followed in the development and testing of this algorithm.

The assumption made by Otsu thresholding, which will be referred to here as standard Otsu thresholding, is that tissue and non-tissue pixels can be separated by their grey-scale values, which is not the case when artefacts are present. However, our method, which will be referred to as the H&E Otsu method and is described in Equation 1 and Algorithm 1, is based on the assumption that the tissue pixels can be identified by being *both* more red than green and more blue than green as compared to non-tissue pixels. The advantage of our method is that all shades of grey have approximately the same value in the red channel as the green channel, so their difference is 0, while pixels of H&E-stained tissue have higher values in the red and blue channels than the green. Setting all negative values in both representations to zero ensures that artefacts with high green channels compared to blue or red channels do not adversely influence the threshold calculation, and are thus considered as background. Only the most “purple-pink” pixels are selected, so pen marks (which are often black, blue, green or red) are also excluded, independent of the pixel’s light intensity.

We compared the performance of H&E Otsu thresholding and standard Otsu thresholding by applying both methods to a dataset of WSIs and assessing the resulting tissue segmentations qualitatively.

Algorithm 1: The H&E Otsu Method

```

RGB Image:  $I \leftarrow [I_R, I_G, I_B]$ 
R - G Representation:  $I_{R-G} \leftarrow I_R - I_G$ 
B - G Representation:  $I_{B-G} \leftarrow I_B - I_G$ 
Tissue Representation:
 $T \leftarrow \text{ReLU}[I_{R-G} \odot I_{B-G}]$ 
Otsu threshold:  $\gamma \leftarrow \text{Otsu}[T]$ 
if  $T[p] > \gamma$  then
| Pixel  $p$  is segmented as tissue
else
| Pixel  $p$  is rejected
end

```

2.1 Data

To compare the performances of the standard Otsu and H&E Otsu methods, we applied both methods to a selection of 60 WSIs of H&E stained duodenal biopsies. Of the 60 WSIs selected:

- 15 contained pen marks
- 15 contained scanning artefacts
- 30 contained no significant artefacts

The 30 WSIs containing artefacts were hand-picked from an available dataset of approximately 4,400 WSIs. We included WSIs containing a wide range of artefacts of different types, shapes and colours. The WSIs were scanned with a wide range of digital scanners (Ventana, Aperio, Hamamatsu and Philips), and the 30 WSIs with no significant artefacts were selected at random from our dataset and matched for scanner type of the 30 WSIs with pen marks or artefact.

2.2 Ethical Statement

All fully anonymized slide scans (and patient data) were obtained with full ethical approval from the Oxfordshire Research Ethics Committee A (IRAS: 162057; PI: Prof. E. Soilleux), and the method was performed in accordance with their guidelines and regulations. Informed consent was obtained from all subjects and/or their legal guardian(s).

3 Results

The standard Otsu and H&E Otsu methods were used to segment the tissue from the 60 WSIs described above. Examples of the WSIs selected and the tissue segmentation of both methods can be viewed in Figure 2. Examples of the tissue segmentation masks generated from both methods are displayed in Supplementary Material. The tissue segmentations were assessed by a single observer, and considered “successful” if all the following were true:

- All tissue was segmented
- All background was rejected from the segmentation
- All bounding boxes were rejected from the segmentation
- All artefacts were rejected from the segmentation

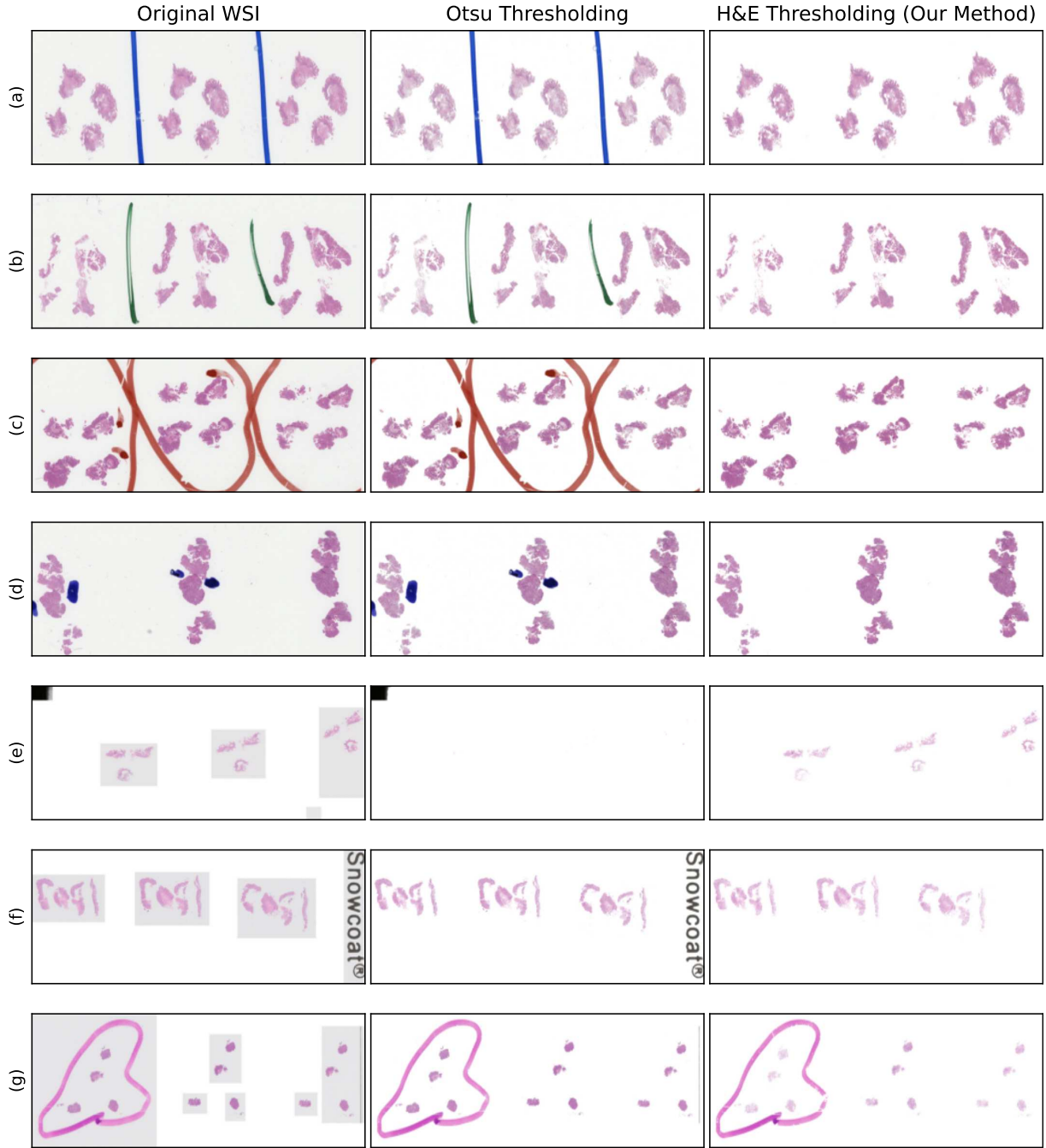


Fig. 2: Seven WSIs of H&E-stained biopsies containing artefacts of a wide range of types and colours. The aim was to segment the tissue without including background and artefacts. Left: The original WSI. Middle: The tissue segmentation from standard Otsu thresholding placed on a white background. Otsu thresholding failed to reject a single artefact and failed to segment the tissue in (e). Right: The tissue segmentation from H&E Otsu thresholding (our method) placed on a white background. H&E Otsu thresholding successfully segmented all tissue and rejected all background and artefacts except the pen marks in (g). H&E Otsu thresholding failed to reject the pen marks in (g) because H&E Otsu thresholding assumes that tissue and non-tissue pixels can be separated through their “pinkness”, which is not true when the pen marks are also pink. Masks for all segmentations can be found in Supplementary Material.

Standard Otsu thresholding rejected pen and scanning artefacts from the tissue segmentation in 0/30 WSIs containing artefacts. In 2/30 WSIs containing artefacts, the influence the artefacts had on the threshold was so great that the tissue was not segmented as tissue (see Figure 2e).

Our method segmented the tissue in all 60/60 WSIs and rejected all artefacts in 29/30 WSIs containing pen and scanning artefacts. The only WSI where pen marks were included in the tissue segmentation can be seen in Figure 2g. Here H&E Otsu thresholding failed to reject the pen marks because tissue and non-tissue pixels could not be separated through their “pinkness”, when the pen marks were also pink.

4 Discussion

H&E Otsu thresholding significantly outperformed standard Otsu thresholding in segmenting tissue from WSIs of H&E-stained biopsies. While standard Otsu thresholding segmented the tissue in all artefact-free WSIs and most WSIs with artefacts, it identified all artefacts as tissue as well. In 2 out of 30 WSIs with artefacts, the presence of artefacts caused the threshold to ignore tissue and include background in the tissue segmentation as seen in Figure 2e.

H&E Otsu thresholding, on the other hand, segmented the tissue in all WSIs and rejected artefacts in all WSIs containing artefacts but one. The only exception can be seen in Figure 2g, which contained pink pen marks that caused both standard Otsu and H&E Otsu methods to fail.

The H&E Otsu method is a rapid, reliable and easily implementable tissue segmentation and artefact removal tool for WSIs of H&E-stained biopsies. In machine learning tasks especially, this tool can be used as a preprocessing step that ensures artefacts do not cause the machine learning algorithm to train on irrelevant patches or patches that contain data leaking pen marks.

It should be noted that the H&E Otsu method is built to segment H&E-stained tissue only, and will not perform as intended on tissue which has been stained with stains that do not appear pink/purple. However, the H&E Otsu method should be relatively simple to generalize to other stains by using representations of the WSI data that uniquely differentiate the stained tissue from

background and artefacts, and will be studied in future research.

Funding

This work was supported by the Pathological Society [PKAG/924] and GlaxoSmithKline [LEAG/781]

5 Author Contributions

B. A. Schreiber devised the H&E Otsu method and wrote the manuscript. J. Denholm and F. Jaekle independently tested and compared standard Otsu and H&E Otsu methods. Histological expertise was provided by M. J. Arends and E. J. Soilleux. The project was initialized by E. J. Soilleux and supervised by K. M. Branson, C.-B. Schönlieb and E. J. Soilleux. All authors were given the opportunity to review and comment on the manuscript.

6 Data Availability

The datasets of WSIs analysed during this current study have not publicly available due to large size of the WSIs and legal considerations. However, low-level representations of the WSIs used in the study have been made available at https://gitlab.developers.cam.ac.uk/bas43/h_and_e_otsu_thresholding.

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Supplementary Files

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- [supplementarymaterial.pdf](#)